

Engineering Challenges & Solutions:

The path to creating our bot was not easy. Many challenges had encumbered us on the way. Here are the most significant challenges out of them all:

1. SSH via wifi on VScode was proving to be impossible, even after hours and hours of trying different solutions. Without SSH, our code cannot be tested, as most of the libraries utilised by the code could only work on the raspberry pi.
 - a. To fix it, we completely ditched using wifi, and instead used ethernet.
 - b. Later we discovered that using a hotspot configured to WPA2 on an older phone was the only way to allow our raspberry pi zero w to connect, though SSH still required a direct ethernet connection to a laptop.
2. Our obstacle detection algorithm could detect obstacles, but it was slow, reported a low fps, struggled with lighting, overloaded our test raspberry pi 3 cpu that was much stronger than the zero w, and drew bounding boxes above obstacles
 - a. To fix it, we changed the algorithm to draw bounding boxes based upon the centre of an obstacle, rather than the corner.